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L. Zewail's Substring Symphony

time limit per test: 3 seconds
memory limit per test: 256 megabytes

Dr. Ahmed Zewail, the Egyptian Nobel laureate and pioneer of femtochemistry, was not only passionate about science but also deeply fascinated by the beauty of mathematical logic and computational challenges. During one of his reflective moments, Dr. Zewail encountered an intriguing problem that captivated his analytical mind:

Given string S of N digits. You will process Q queries. In the i-th query, you take a substring from l_i to r_i from the original string, sort this substring, and write the whole string with this sorted substring on a board. You have to calculate the number of different strings after performing all queries.

At first glance, the problem seemed like a simple string manipulation task, but Dr. Zewail quickly realized that maintaining the integrity of the original string while applying sorted modifications required a delicate balance of data structures and optimization. Much like his experiments capturing molecules in femtoseconds, every operation had to be precise and efficient. This blend of logical rigor and creative problem-solving reminded him that whether in chemistry or com

Input

First line contains two integers $N(1 \leq N \leq 2 \times 10^5)$, the length of the string, $Q(1 \leq Q \leq 2 \times 10^5)$.

Second line contains string S consisting of Digits $(0 \leq s_i \leq 9)$.

Next Q lines, the $i-th$ line contains two integers l_i, r_i $(1 \leq l_i \leq r_i \leq N)$.

Output

Output a single integer, the answer to the problem.

Example

input	Copy
8 5 19708041 1 3 4 5 4 6 5 6 3 8	

output	Copy
4	

Note

Testcase explanation: S = "19708041".

1. after sorting $S[1:3]$, S = 17908041
2. after sorting $S[4:5]$, S = 19708041
3. after sorting $S[4:6]$, S = 19700841
4. after sorting $S[5:6]$, S = 19700841
5. after sorting $S[3:8]$, S = 19001478

There are 4 distinct strings $[17908041, 19708041, 19700841, 19001478]$

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